

Chetna Pal

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Education

Integrated MTech in Computational and Data Science
VIT Bhopal University | CGPA: 8.21/10 | Expected May 2027

Projects

- SafeSphere – Community Safety Risk Modeling System
 1. Designed a CNN-LSTM model leveraging geospatial and temporal data to assess commuter safety risk, reaching 87% accuracy in predicting high-risk zones during peak hours and providing actionable insights.
 2. Engineered an interactive, map-based risk visualization frontend using Folium and connected it to a modular FastAPI + Streamlit architecture; system now used by 15+ analysts.
 3. Improved risk prediction accuracy by analyzing location, time of day, and incident trends.

SKILLS

Technical Skills

- Programming Languages: Python, C++, SQL
- Machine Learning: Scikit-learn, CNN, LSTM, Model Evaluation
- Backend Development: FastAPI, REST APIs
- Data Analysis: Pandas, NumPy
- Data Visualization: Streamlit, Folium
- Databases: MySQL
- Cloud Technologies: AWS (EC2, S3, IAM – basics)
- Core Concepts: Data Structures, Algorithms, DBMS, OS, Computer Networks
- Tools: Git, Jupyter Notebook, VS Code

RELEVANT COURSEWORK

- Machine Learning
- Data Structures and Algorithms
- Database Management Systems
- Operating Systems
- Computer Networks

CERTIFICATIONS

- Simulink Onramp by Mathworks
- Matlab by Mathworks
- Introducing Generative AI with AWS by Udacity
- Applied Machine Learning in Python by University of Michigan
- MERN Full stack with AI Internship Program by Ethnus
- The Ultimate Job Ready Data Science Course by Code With Harry

EXTRA-CURRICIAL

- Active member of the **Data Science Club**; volunteered in technical events and hosted sessions to engage participants.
- Led and organized **community teaching initiatives** for children of construction workers, focusing on basic literacy and numeracy.
- Participated in multiple **musical performances and talent hunts**, demonstrating teamwork, creativity, and stage presence.
- Lead dancer in official **VIT Bhopal ad shoot (2024)**.

SUMMARY

MTech student specializing in Computational and Data Science with hands-on experience in machine learning, data analysis, and backend development. Proficient in Python, FastAPI, AWS, SQL, and data visualization tools. Experienced in building real-world AI-driven projects with strong problem-solving and analytical skills.

